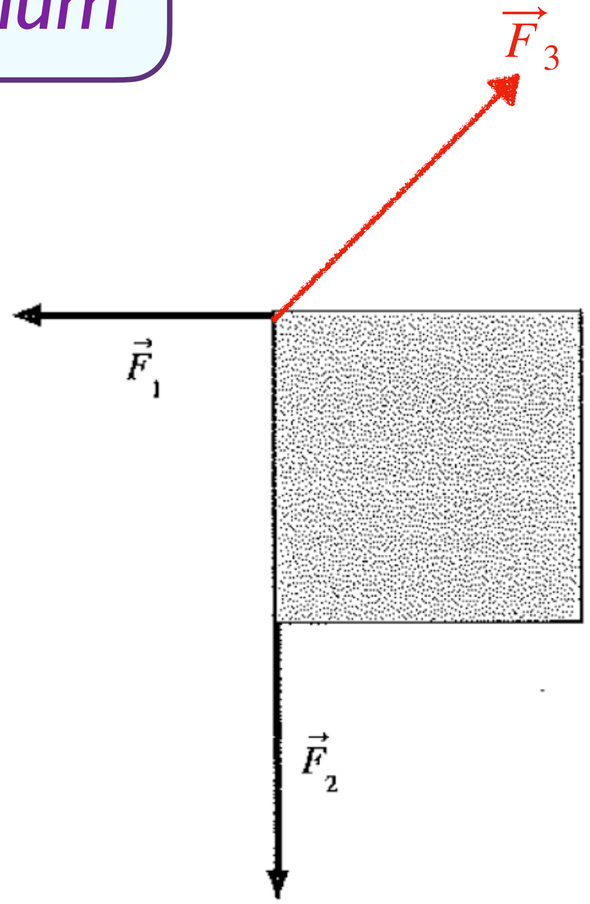
Discussion Exercise 7

University of California, Riverside
July 17, 2023

TA: Yifan Liu

1. Torque and Static Equilibrium

Forces $\vec{F}_1$ and $\vec{F}_2$ have the same magnitude. They are applied to the corners of a square plate as shown. Draw and label a *single* force vector $\vec{F}_3$ applied to the appropriate point on the plate that will cause the plate to be in total static equilibrium. Make sure the length of your vector shows the magnitude of $\vec{F}_3$ relative to $\vec{F}_1$ and $\vec{F}_2$.



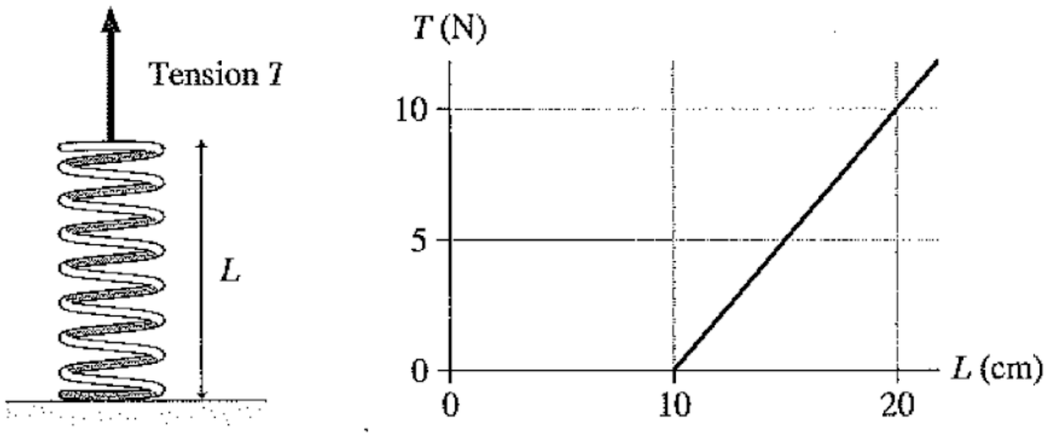
The equilibrium conditions:

1. The net torque is zero: $\sum_i \vec{\tau}_i = 0$

2. The net force is zero: $\sum_i \vec{F}_i = 0$

2. Springs and Hooke's Law

A spring is attached to the floor and pulled straight up by a string. The spring's tension is measured. The graph shows the tension in the spring as a function of the spring's length L.



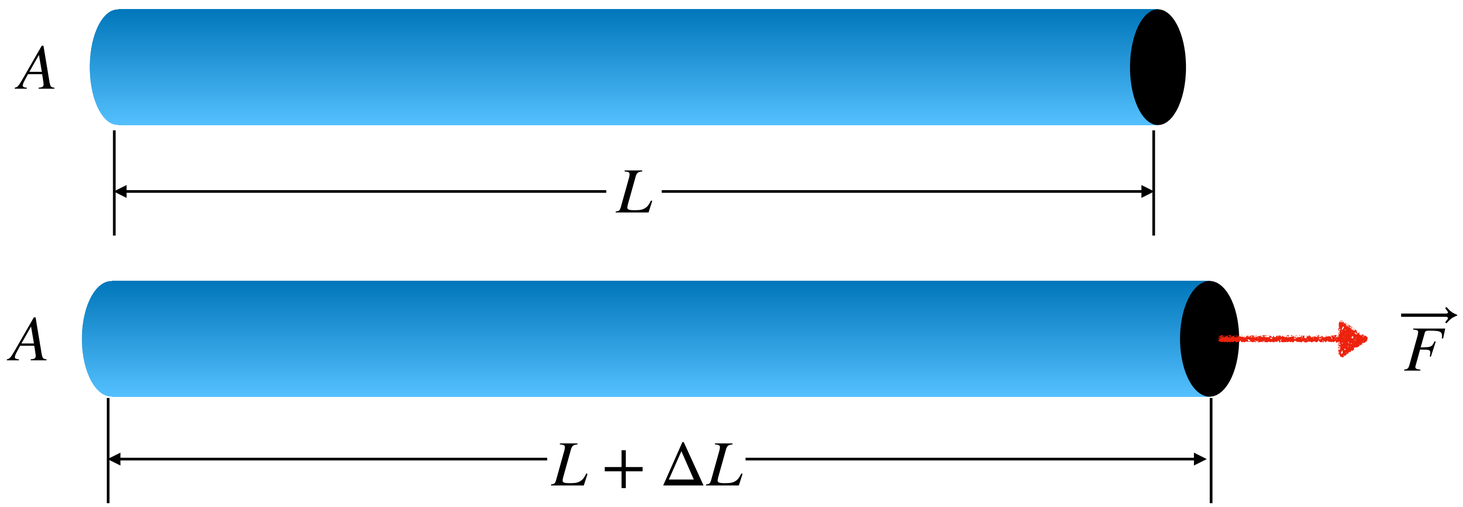
a. Does this spring obey Hooke's Law? Explain why or why not.

Yes. Based on Hooke's Law $T = k\Delta L$, the tension force is a linear function of change in the length/distance, which can be directly seen from the graph.

b. If it does, what is the spring constant?

Hooke's Law $T = k\Delta L$ implies that $k = T/\Delta L$, i.e. we may interpret the spring constant as the slope of T v.s L graph! $k = 100N/m$

3. Stretching and Compressing Materials



Another version of Hooke's Law: $\frac{F}{A} = E\frac{\Delta L}{L}$, where F is the force applied, A is the cross-section area, E is the modulus of elasticity depending on material,L is the length of the wire, ΔL is change in length due to F

3. Stretching and Compressing Materials

A force stretches a wire by 1 mm.

1. A second wire of the same materials has the same cross section and twice the length. How far will it be stretched by the same force? Explain.

Notice that the second wire has the same material (same E), the same force F, the same A. Thus we may rewrite the equation $F/A = E\Delta L/L$ as $\Delta L = \left(\frac{F}{AE}\right) L$, in which the variable inside the parenthesis can be treated as a constant. Therefore, $\Delta L \propto L$ and ΔL for the second wire should be 2mm since the length is doubled.

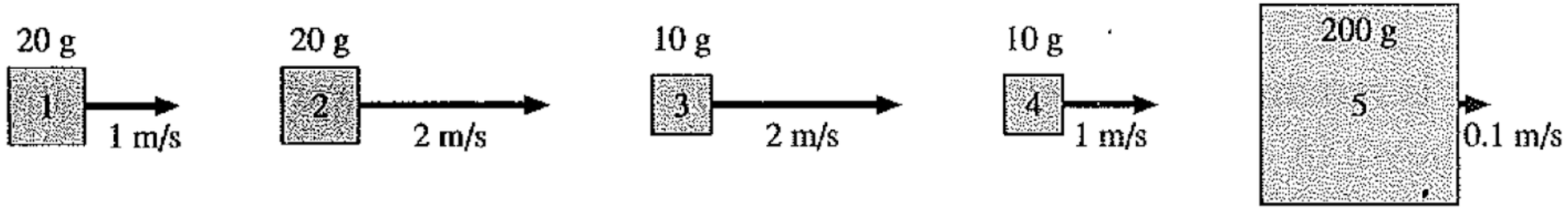
2. A third wire of the same material has the same length but twice the diameter of the first wire. How far will it be stretched by the same force? Explain.

Notice that the third wire has the same material (same E), the same force F, the same L. Thus we may rewrite the equation $F/A = E\Delta L/L$ as $\Delta L = \left(\frac{FL}{E}\right) \frac{1}{A}$, in which the variable inside the parenthesis can be treated as a constant. Therefore, $\Delta L \propto A^{-1}$ and ΔL for the third wire should be 0.25mm since the area is quadrupled.

4. Momentum

$$\vec{p} = m\vec{v}$$

Rank in order, from largest to smallest, the momenta $(p_x)_1$ to $(p_x)_5$.



$p_1 = m_1 v_1 = 0.02\ kg \cdot m/s$

$p_2 = 0.04\ kg \cdot m/s$

$p_3 = 0.02\ kg \cdot m/s$

$p_4 = 0.01\ kg \cdot m/s$

$p_5 = 0.02\ kg \cdot m/s$

Therefore, $p_2 > p_5 = p_3 = p_1 > p_4$